Predicting Building - Level Flood Risk by Analysis Comparison of Multi-Layer Perceptron and Bayesian Logistic Regression

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Introduction & Problem

•**The Problem:** Current FEMA Flood Insurance Rate Maps (FIRMs) rely on macro-level topographic data, largely ignoring specific physical characteristics of individual structures.

•**The Consequence:** Two houses in the exact same flood zone can have very different actual risk levels depending on micro-level factors (like foundation height).

•**Our Objective:** Leverage National Structure Inventory (NSI) data to model flood risk at the individual building level, moving from zone-level mapping to building-level prediction

Data Overview

Dataset: National Structure Inventory (NSI).

FROM GPKG TO CSV, extract the 153K rows of buildings with FIRM info

Target Variable: FEMA FIRM Zones.

•**High-Risk (1):** VE, AE, A, AO, AH.

•**Low-Risk (0):** Area X.

•**Key Features:** Ground elevation, foundation height, structural value, year built, square footage, number of stories, building type, and foundation type.

•Download and documentation:

https://www.hec.usace.army.mil/confluence/nsi/technicalreferences/latest/technical-documentation

References

Demissie, Z., et al. (2024). *Applied Computing and Geosciences, 23*, 100183.

Chen, Y., et al. (2023). *Frontiers in Earth Science, 11*, 1117004.

Ouma, Y. O., & Lawrence, O. (2023). *Int. J. of Intelligent Systems 2023*, 5672401.

Qin, X., et al. (2025). *npj Urban Sustainability, 5*, 19.

U.S. Army Corps of Engineers. *National Structure Inventory (NSI) Technical Documentation.*

Methodology

1. Custom MLP (By Lang)

•**Implementation:** It was built manually by using Python and NumPy matrix operations

Architecture: * Input Layer: d dimensional feature space.

$$h = \max(0, XW_1 + b_1)$$

- Hidden Layer: 16 nodes with **ReLU** activation $\hat{y} = \sigma(hW_2 + b_2) = \frac{1}{1 + e^{-(hW_2 + b_2)}}$
- Output Layer: 1 node with **Sigmoid** activation for probability mapping

•**Optimization:** Full-batch Gradient Descent minimizing Binary Cross-Entropy (BCE) loss (eta=0.05, 2000 epochs) : $J = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$

2. Bayesian Logistic Regression(By Jiada)

•**Implementation:** built using R and rstanarm library

•**Architecture:** * Input: d dimensional feature space.

- Prior: Normal distribution with mean=0, std= 2.5 for features. Mean=0, std=10 for intercept $w \sim N_p(w^{(0)}, \Sigma_0)$
- Likelihood: Follow a Bernoulli distribution $y_i | w, x_i \sim Bern(\theta = \sigma(w^T x_i))$

•**Optimization:** Use MCMC sampling to retrieve feature weights with iter=1000, chain=4, warmup=500,

Results & Interpretation

1. MLP Results

•**Predictive Metrics:** * Accuracy: **93.97%**

- Precision: 0.9243
- Recall: **0.9440**
- F1-Score: 0.9341

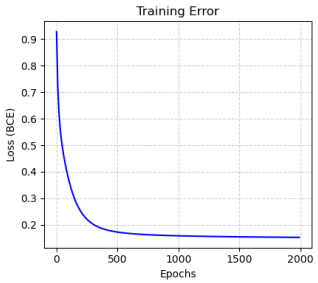
2. Bayesian Logistic Regression Results

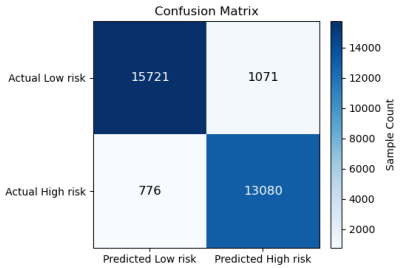
•**Predictive Metrics:** * Accuracy: 92.52%

- Precision: 0.8890
- Recall: 0.7675
- F1-Score: 0.8238

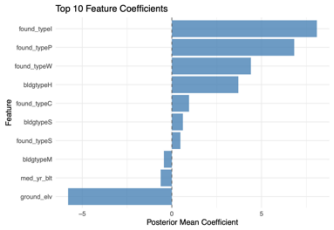
•**Key Takeaway:** The MLP outperforms the Bayesian Logistic Regression across all metrics, achieving higher accuracy (93.97% vs 92.52%), recall (0.9440 vs 0.7675), and F1-score (0.9341 vs 0.8238). The exceptionally high Recall indicates a very low rate of False Negatives. This ensures vulnerable structures are not mistakenly classified as safe, which is paramount for municipal planning.

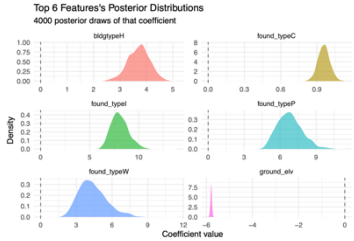
Figures of MLP Results





Figures of BLR Results





Conclusions & Future Work

Model Trade-offs: The MLP did better at capturing non-linear patterns, but it's a black-box··· unlike the BLR which gives interpretable coefficients. The BLR trades accuracy for interpretability. Its coefficients show foundation type and ground elevation drive flood risk the most.

Future Directions: * Expand the dataset to include diverse geographical regions.

•Incorporate a post-hoc interpretability framework, such as **SHAP values**, to understand the marginal contribution of individual building attributes (e.g., ground elevation) to the MLP's predictions. The BLR model has higher interpretability of the features and uncertainty. The combination of a neural network and a Bayesian model (BNN) could improve the predictability.